

1. Database Is a Structured system for storing and organising data so it can be easily accessed.

* Database are the foundation of most modern technology without them, apps, websites and services would fail to deliver personalised and seamless experience.

→ CRMs utilize databases to store customer information.

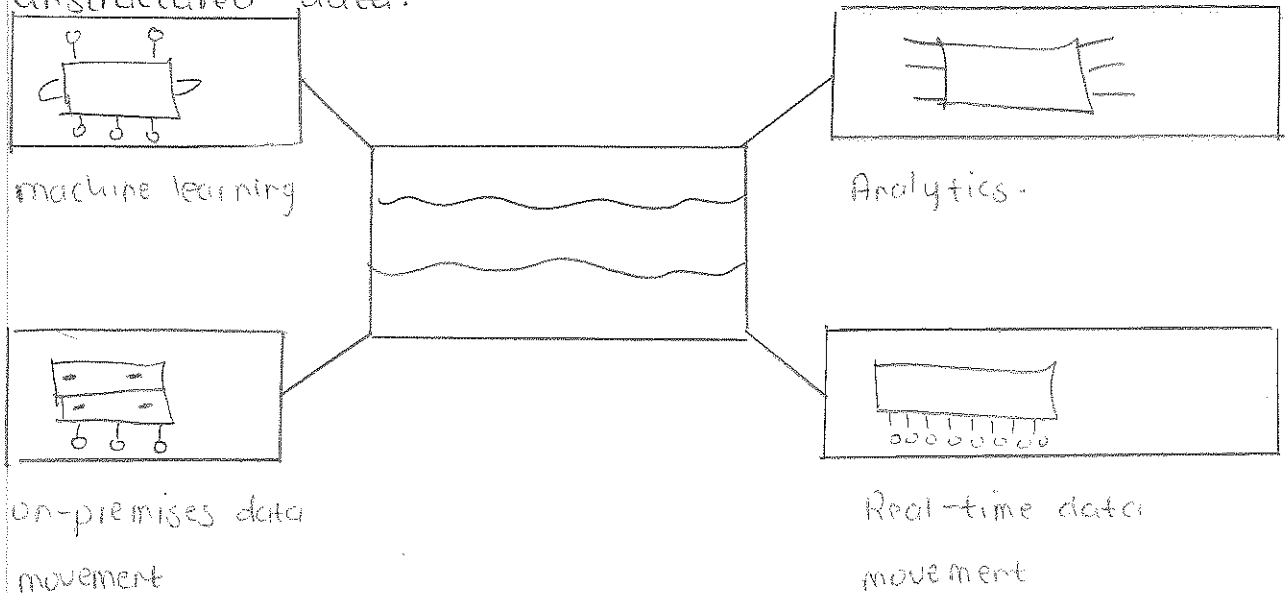
→ Electronic Health Records (EHRs) use data to store and retrieve patient information..

2. Data Warehouse Is a Centralized repository that stores Structured data (Excell, Sheet) and Semi-structured data (webpages)

* A database stores the current data required to power an application whereas a data warehouse stores current and historical data for one or more systems in a predefined and fixed schema for the purpose of analyzing data.

→ Companies use data warehouse to maintain historical data records.

3. Data lake Is a Centralized repository designed to store, process and secure large amounts of structured, semistructured and unstructured data.



* Data lake store all types of raw data, which data scientists may use for a variety of projects while data warehouse store cleaned and processed data, which can be used to source analytic or operational reporting.

→ when flexibility, cost efficient storage, and support for diverse or structured data are essential. But choose data warehouse when you need fast, reliable access to curated data.

4. Data Mart Is a Subset of a data warehouse focused on a particular line of business, department or Subject area.

* Data marts have a single-subject focus and are more decentralised in nature compared to data warehouse which typically store data from multiple business units.

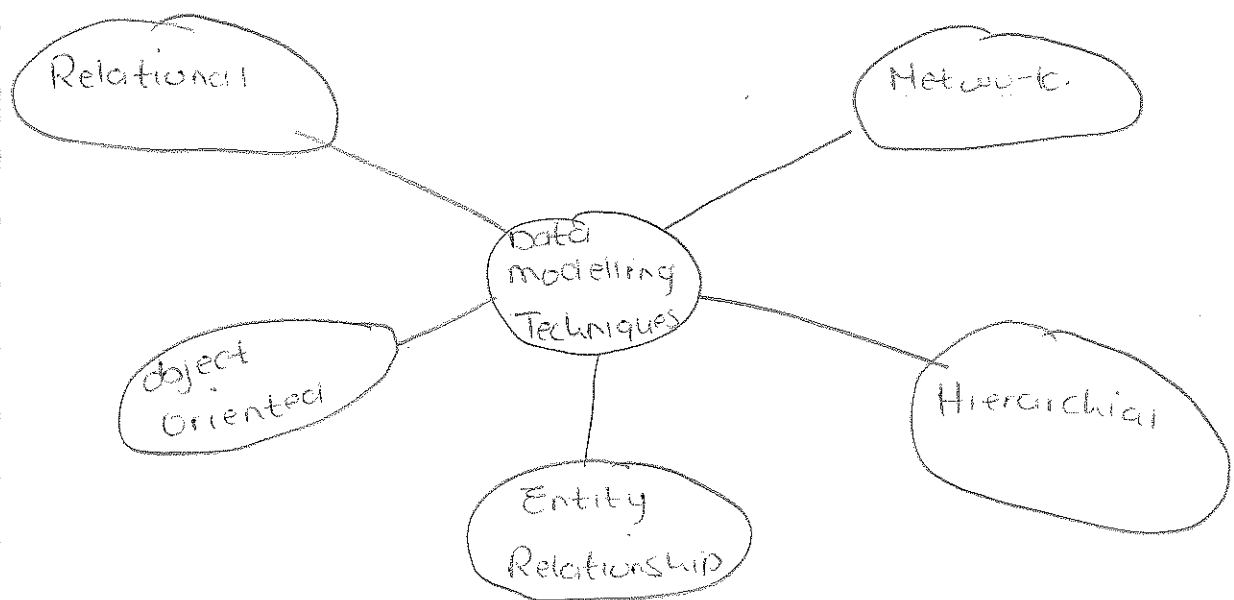
* Marketing team may use data marts to analyze consumer behaviours.

5. Structured data Is a data whose elements are addressed for effective analysis e.g. Relational data.

Semi-Structured data Is information that does not reside in a relational database but has some organizational properties that make it easier to analyze e.g. XML data.

UnStructured data Is data which is not organised in a predefined manner or does not have a predefined data model e.g. Word.

6. Data Modelling Is the process of analyzing and defining all the different data types your business collects and produces.



It is important because you can document what type of data you have, how you use it and the data management requirements surrounding its usage, protection and governance.

→ Conceptual data model - This is a model used in the requirements-gathering process.

→ Representational data model - It is used to represent only the logical part of the database and does not represent the physical structure of database.

→ physical data model - It is used to practically implement Relational data model.

1. Data mining is the process of finding anomalies, patterns and correlations within large data sets to predict outcomes, * It is used to transform large data sets into actionable business insights.

→ Banks use data mining to better understand market risks.

→ Supermarkets use joint purchasing patterns to identify products.

8. Database management Is a software system that allows users to create, define, manipulate and manage database.

* mySQL is a free, open source relational database management system.

* Oracle - Is a Commercial, multi-model database management system.

* PostgreSQL - Is a free-open source relational database management.

* MongoDB - Is an open source, NoSQL, document-oriented database management system.

9. Extract, transform load Is a three-step process that retrieves data, transform and improves it and delivers the Consolidated data into a data warehouse. The ETL process Consolidates data, enabling businesses to analyse it effectively and deliver - Insights.

10. Use CSV for Simple tabular data.

- JSON for web APIs and nested data.

- Parquet for big data analytics and storage efficiency.

- Avro used in big data ecosystems like Hadoop and Apache Kafka.

#. Section B - Artificial Intelligence & Machine Learning.

11. AI Is a Set of technologies that empowers Computers to learn, reason and perform a variety of advanced tasks in ways that used to require human intelligence.

* Narrow AI Systems are Confined to the tasks they were designed for, while General AI would theoretically be able to think, reason and learn as a human does.

14. Natural Language processing is technology that allows computers to Interpret, manipulate, and Comprehend human language.

- * Auto Correct and Spell Check.
- * Question Answering
- * Machine Translation.

15. Large language model is a type of artificial Intelligence (AI) program that can recognise and generate text among other tasks.

- * Customer Service - large language models provide quick and precise answers to queries from users.
- * Content Creation - They facilitate writing, Summary of text and automatic translation.

16. Generative AI Refers to the use of AI to create new content like text, Images, audio and videos.

→ It is powered by foundation models (large AI models) that can multi-task and perform out of the box tasks.

- * Chat GPT - for versatility
- * Microsoft power apps - for in-depth app development.
- * Adobe photoshop - for AI-generated Image enhancement.

17. Training Dataset Refers to the database used to create a representative model by Inferring from the data.

- * High quality data is essential because AI models learn from examples therefore accurate, diverse and clean data directly determines the models reliability, accuracy and fairness. poor quality data leads to "garbage in, garbage out" Scenarios, resulting in biased, inaccurate and Unreliable predictions that can fail in real-world.

* Narrow AI - focuses on a single task and is restricted from moving beyond that task to solve unknown problems. General AI can solve many problems using human-like intelligence.

12. Machine learning is a way to train software, called a model, to make predictions or generate content using data.

* Traditional programming is more fixed approach where the programmer designs the solution explicitly, while ML is a more flexible and adaptive approach where ML ^{models} ~~is a more~~ learns from data to generate a solution.

→ Supervised learning - Trains a machine learning algorithm on labelled data sets, effectively giving it a key to understand how to interpret the information it's being fed.

→ Unsupervised learning - Trains the algorithm on unlabeled data, or data without tags.

→ Reinforcement learning - Is the closest type of machine learning to how humans actually learn.

13. Deep learning is a subset of machine learning driven by multilayered neural networks whose designs is inspired by the structure of the human brain.

* Deep learning is a subfield of machine learning, machine learning is a subset of AI and AI is the overarching system.

* Neural Networks are machine learning algorithm inspired by the human brain.

* Occurs by creating connections and adjusting the weights of connections between neurons through a process called training.

- JSON is commonly used when working with API and data exchange because it does not use bulky closing tags.

Example of a JSON object

- "First Name": Vutlhari
- "Last Name": Kubayi

23. Prompt Engineering is the process of designing, refining and optimising inputs (prompts) to guide generative AI models towards producing accurate, relevant and high quality outputs.

- It is an important skill because it transforms AI tools from generic, conversational bots into precise high-performance thinking partners.

Examples

- Event foster
- Restaurant menu

24. A MAP function is a high level tool used in programming, spreadsheet and mathematics to transform data.

- In programming and data processing MAP function utilise code, algorithms and logical structures to automate these transformations, converting unorganised inputs into structured outputs like reports, graphs or actionable insights.

Example in Java Script

* `Console.log ("Hello, world!")`

25. Cloud Computing is the on-demand delivery of IT resources such as computing power, storage and data bases over the internet with pay-as-you-go pricing.

19. API is a set of rules and protocols that allow different software programs to communicate with each other

- APIs are important because they act as secure, standardised intermediaries that allow different systems to communicate, share data and utilise functionality without direct integration

20. An API key is a unique alphanumeric string used to identify and authenticate an application or project making a request to an API.

- An API key purpose is to serve as an intermediary that allows two different software programs to communicate and exchange data with each other.

How it should be stored	If exposed publicly
- Use environment variables - Apply the principle of least privilege - Avoid Client side exposure	- Financial loss - Legal and Regulatory Penalties - Service disruption

21. A REST API is an architectural style for designing networked applications that allows different systems to communicate over the internet.

- REST stands for Representational State Transfer.

* GET - used to retrieve a representation of a resource.

* POST - Used to submit data to a server to create a new subordinate resource.

* PUT - Used to replace an entire existing resource with new data or create a resource at a specific URL if it doesn't exist

* DELETE - used to permanently remove a specific resource from the server

22. JSON is a light weight, text based data interchange format used to store and transmit structured data primarily between a server and a web application.

Four Libraries commonly used

- NumPy : Provides support for large multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on these data structures.
- Pandas : Offers data structures and data analysis tools, making it easier to manipulate structured data, such as tables and time series.
- Matplotlib : A plotting library used for creating static, animated and interactive visualisation in Python

26 Version Control System is software that tracks and manages changes to files - primarily source code - over time, creating a recorded history of modifications.

Collaboration without Conflicts: multiple analysts can work on the same projects simultaneously. Git's merging capabilities combine their different contributions while identifying exactly where changes might conflict.

27 Jupyter Notebook is an open-source web-based application designed for creating and sharing interactive computational documents. It allows users to combine live code, visualisations and narrative text in a single document, making it highly useful for data analysis, scientific computing and machine learning workflows.

- It is mostly used with the Python programming language.
- The popularity of Jupyter Notebooks among data analysts and data scientists stems from its ability to facilitate interactive and iterative data exploration, visualisation and reporting.

28. Python is a high-level, interpreted, and object-oriented programming language known for its simplicity, readability and versatility. It emphasises clear syntax and dynamic semantics making it accessible for both beginners and experienced developers.

- Its widespread use in data analysis, artificial intelligence (AI), machine learning, web development, automation and scientific computing explains why it is considered one of the most popular languages for data analysts and AI.

31. Data governance is a structured framework that enables organizations to manage their data assets effectively throughout their lifecycle.

→ It involves establishing policies, roles, processes and standards to ensure data is accurate, consistent, secure, available, and used responsibly.

32. Bias in AI refers to systematic and repeatable tendencies within an AI system to produce unfair or discriminatory outcomes often favoring certain groups over others based on characteristics like race, gender or ethnicity.

* The consequences of bias in AI can be significant, including legal liabilities, damage to a company's reputation, and the exclusion of certain groups from opportunities or services.

33. A data analyst focuses on interpreting data to extract meaningful insights, often using tools like SQL, Excel and visualization software such as Tableau or Power BI.

A data scientist builds predictive models and advanced algorithms to analyze complex data sets, requiring skills in statistical analysis, machine learning, and programming languages like Python or R.

A data engineer designs, constructs, and maintains the infrastructure—such as data pipelines and databases that allow data collection, storage, and processing at scale.

An AI engineer (or machine learning engineer) specializes in deploying AI models into production environments, creating ~~build~~ scalable AI systems, and working with frameworks like TensorFlow or PyTorch.

29. Data privacy is the proper handling, processing and protection of Individuals personal Information to ensure it is not misused, accessed without authorization, or exposed to risks.

* GDPR is a Comprehensive data protection law enacted by the European Union that came into effect on May 25, 2018.

30. Data ethics Is the moral principles that guide the the responsible Collection analysis, and use of data and AI technologies.

→ Using AI for discriminatory hiring practices! When AI algorithms are trained on biased data, they can perpetuate discrimination against certain groups, such as minorities or women, in hiring decisions. This is problematic because it reinforces societal inequalities.

→ Invasive data collection without consent! Collecting personal data without explicit user consent, such as tracking online behaviour or biometric data, breaches privacy rights and erodes trust. This practice is problematic because it disregards individual autonomy and can lead to misuse or unauthorized sharing of sensitive information..

→ Bias in AI decision-making system! AI system used in Criminal justice or credit scoring that produces biased outcomes can unfairly disadvantage certain populations. This is problematic because it can cause social harm, violate fairness principles and lead to unjust treatment of individuals.

REFERENCES.

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34. Business Intelligence Refers to the Strategies, technologies and processes that organization use to collect, manage, analyze and interpret data to support better decision-making.

* Looker provides a unified view of a company's data facilitating analysis across different datasets and supporting strategic decisions through visualizations and reports.

* Qlik offers a comprehensive platform for collecting, analyzing and presenting business data, helping organisations interpret large data sets and identify actionable insights.

35. Artificial Intelligence and machine learning.

Because it is impossible to go about a day without using or hearing about AI, with AI, a person can always multi-task and it is opening up a world of opportunities.

36. APIs, Tools & Developer Concepts:

→ AI made

- Google

- YouTube.

37. Data and AI are shifting roles from manual execution to oversight and strategic interpretation, forcing a move towards continuous reskilling to remain relevant.